

Akash Karthik

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EDUCATION

Bachelor of Science, Computer Science

Expected May 2027

San Jose State University, San Jose, CA

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Computer Systems, Intro to Circuits

TECHNICAL SKILLS

Languages: Python, C/C++, Java, TypeScript/JavaScript, SQL

Backend & APIs: FastAPI, Flask, Node.js/Express, REST/JSON, WebSockets, real-time streaming, LLM integration, schema design

Systems: Real-time and deterministic execution, fixed-point arithmetic, embedded firmware, concurrency, performance profiling

Testing & Deployment: pytest, Jest, GitHub Actions, Docker, structured logging, Git/GitHub, Postman, React

EXPERIENCE

Software Lead

Jun 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

- Lead software planning for Python tooling, controller validation, diagnostics, and shared testing workflows across five project teams
- Standardize project requirements, test evidence, and technical reviews spanning dashboard, simulation, and vehicle software

Software Engineer Intern

Feb 2026 - Present

Argus Defense | San Jose, CA

- Developed and validated flight-control software in C on PX4, using structured telemetry logs to reproduce issues and evaluate changes
- Integrated visual-inertial odometry and iterated configuration through repeatable tests and documented results

Software Controls Engineer

Jul 2025 - Jun 2026

Spartan Racing, Formula SAE | San Jose, CA

- Implemented a real-time control routine in C using fixed-point arithmetic, saturation, and rate limiting to meet hard deterministic timing budgets on an embedded target
- Built Python parsers and a replay harness that validated changes against recorded logs before deployment to hardware

Software Development Intern

Sep 2024 - Oct 2024

Zip Intelligence | Tracy, CA

- Took ownership of a feature from ticket through verified production release, shipping via pull requests with setup and change documentation
- Implemented frontend validation and clearer error states, then paired with engineers to harden REST endpoints for status codes and edge cases
- Added SQL Server update scripts and structured logs to speed reproduction and issue triage

PROJECT EXPERIENCE

Tickr — AI-Coached Paper-Trading Game

2025

TypeScript/JavaScript, React, LLM integration, responsive UI

- Built registration, sign-in, game, and history flows for a scenario-driven trading game with an LLM-backed coach
- Implemented quest and prompt logic, client-side state management, and a mobile-responsive interface

SR-Wireless-CAN — Real-Time Telemetry Platform

2026

Spartan Racing, Formula SAE | San Jose, CA | Backend lead

Python, FastAPI, WebSockets, React, SQLite, Docker Compose

- Led backend development for a real-time telemetry platform, building a FastAPI and WebSocket service that streams live data to a React dashboard with SQLite persistence
- Implemented structured upload and merge of signal definitions with conflict resolution, and a bounded search window keeping large catalogs responsive
- Containerized the backend, frontend, and database with Docker Compose for one-command deployment across machines